Hack The Box, Cyber Apocalypse CTF

The Salt Crown, Digital Forensics

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Category: Digital Forensics

Challenge name : The compressed Truth

Challenge Overview

This challenge provides a folder styled as a captured Windows Registry hive set. The scenario centers on a threat actor, "CROWQUILL," who compromises the workstation of a user named vmarr, dumps credentials from a running KeePass password manager, stages sensitive files, and exfiltrates them as a compressed archive. The goal is to reconstruct CROWQUILL's actions purely from registry artifacts using RegRipper.

Tools Used

Tool	Purpose
RegRipper	Parses Windows Registry hives (e.g. NTUSER.DAT) and runs plugins against known keys to surface forensic artifacts.
7-Zip plugin (RegRipper)	Reads the 7-Zip "PanelHistory"/archive-handling registry keys, which log files and folders opened, browsed, or extracted through 7-Zip's file manager.
Shellbags / folder-access artifacts	Track which folders a user opened in Windows Explorer, including full paths, used to confirm staging locations.
TypedPaths	Registry key that logs paths manually typed into Explorer's address bar.
Copy/compression history	Registry-based history of files copied or compressed via 7-Zip, used to trace movement and packaging of stolen data.

Flag 1. Identifying the Credential-Theft Tool

> ***Question:*** *CROWQUILL did not break the lock, they took the key while it was still held. A tool was brought for one purpose: extract secrets from memory before they could be put away. What is its name?*

Methodology

The provided evidence folder is styled to resemble a captured Windows Registry hive set.

```
┌──(alif㉿kali)-[~/Hack the box CTF]
└─$ ls
C

┌──(alif㉿kali)-[~/Hack the box CTF]
└─$ tree C
C
|-- Users
|   |-- cyberjunkie
|   |   |-- NTUSER.DAT
|   |   |-- ntuser.dat.LOG1
|   |   `-- ntuser.dat.LOG2
|   |-- Default
|   |   |-- NTUSER.DAT
|   |   |-- NTUSER.DAT.LOG1
|   |   `-- NTUSER.DAT.LOG2
|   `-- vmarr
|       |-- NTUSER.DAT
|       `-- ntuser.dat.LOG1
`-- Windows
    `-- System32
        `-- config
            |-- DEFAULT
            |-- DEFAULT.LOG1
            `-- DEFAULT.LOG2

8 directories, 11 files

┌──(alif㉿kali)-[~/Hack the box CTF]
└─$ 
```

The NTUSER.DAT hive belonging to the vmarr user was analyzed with RegRipper, and the results were saved to vmarr_report.txt:

```
┌──(alif㉿kali)-[~/Hack the box CTF]
└─$ cd "../vmarr"
regripper -r NTUSER.DAT -a > vmarr_report.txt
cd: no such file or directory: ../vmarr
```

Reviewing the output of the 7-Zip plugin revealed evidence that KeePass (a credential management application) was installed, alongside signs that KeeFarce, a known tool used to steal credentials from KeePass, had been extracted and run on the system:

```
sevenzip v.20210329
- Gets records of histories from 7-Zip keys

FM LastWrite: [2026-06-18 13:26:47Z]

Compression LastWrite: [2026-06-18 13:25:06Z]

Extraction LastWrite: [2026-06-18 13:15:15Z]

FM\PanelPath0: c:\users\vmarr\desktop\working\

Compression\ArcHistory:
  C:\Users\Public\Pictures\shardchain.tar

Extraction\PathHistory:
  C:\Users\vmarr\AppData\Local\Temp\writ\KeeFarce\


FM\CopyHistory:
  c:\users\public\music\saltwork\
  c:\users\public\music\saltwork


FM\FolderHistory:
  c:\users\vmarr\desktop\working\
  c:\users\public\music\saltwork\
  c:\users\vmarr\appdata\Roaming\KeePass\
  c:\users\vmarr\appdata\Roaming\
  c:\users\vmarr\appdata\
  c:\users\vmarr\appdata\Local\
  C:\Users\vmarr\Documents\Registry\shard_references\
  C:\Users\vmarr\Documents\Registry\
  C:\Users\vmarr\Documents\Registry\shard_storage\
  C:\Users\vmarr\Documents\Registry\shard_storage\ShardKeepass_FirstMark\
  C:\Users\vmarr\Documents\Registry\shard_ref\
  C:\Users\vmarr\Documents\Registry\internal_reports\
  C:\Users\vmarr\Documents\Registry\custody_chains\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\oath_records_cinderbound_vol2\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\oath_records_cinderbound_vol2\saltoaths_s
ecretive\
  C:\Users\vmarr\Documents\
  C:\Users\vmarr\Documents\Personal\
```

KeeFarce works by extracting decrypted credentials directly out of a running KeePass process's memory, before the user locks or closes the database, which matches the wording of the question exactly.

> **Answer:** KeeFarce

Flag 2. Timestamp of Tool Extraction

> ***Question:*** *The Registry keeps time as faithfully as it keeps names. The moment CROWQUILL's tool first touched the system is preserved in the artifact. When was the tool extracted? (YYYY-MM-DD hh:mm:ss)*

Methodology

The same 7-Zip plugin output from Flag 1 also records the last-write time associated with the KeeFarce archive's extraction. This timestamp marks the moment the tool first appeared on the system.

> **Answer:** 2026-06-18 13:15:15

Flag 3. Deepest Enumerated Folder in the Archive

> ***Question:*** *One archive file caught the operative's eye during enumeration, its contents inspected before staging began. What is the deepest folder that was enumerated inside the archive file?*

Methodology

The 7-Zip plugin output shows that oath_records_cinderbound_vol2.zip was opened and browsed before the KeePass database was accessed:

```
----------------------------------------
sevenzip v.20210329
- Gets records of histories from 7-Zip keys

FM LastWrite: [2026-06-18 13:26:47Z]

Compression LastWrite: [2026-06-18 13:25:06Z]

Extraction LastWrite: [2026-06-18 13:15:15Z]

FM\PanelPath0: c:\users\vmarr\desktop\working\

Compression\ArcHistory:
  C:\Users\Public\Pictures\shardchain.tar

Extraction\PathHistory:
  C:\Users\vmarr\AppData\Local\Temp\writ\KeeFarce\


FM\CopyHistory:
  c:\users\public\music\saltwork\
  c:\users\public\music\saltwork


FM\FolderHistory:
  c:\users\vmarr\desktop\working\
  c:\users\public\music\saltwork\
  c:\users\vmarr\appdata\Roaming\KeePass\
  c:\users\vmarr\appdata\Roaming\
  c:\users\vmarr\appdata\
  c:\users\vmarr\appdata\Local\
  C:\Users\vmarr\Documents\Registry\shard_references\
  C:\Users\vmarr\Documents\Registry\
  C:\Users\vmarr\Documents\Registry\shard_storage\
  C:\Users\vmarr\Documents\Registry\shard_storage\ShardKeepass_FirstMark\
  C:\Users\vmarr\Documents\Registry\shard_ref\
  C:\Users\vmarr\Documents\Registry\internal_reports\
  C:\Users\vmarr\Documents\Registry\custody_chains\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\oath_records_cinderbound_vol2\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\oath_records_cinderbound_vol2\saltoaths_secret
ive\
  C:\Users\vmarr\Documents\
  C:\Users\vmarr\Documents\Personal\
  C:\Users\vmarr\
  C:\Users\vmarr\Downloads\
  C:\Users\
  C:\
  Computer\
```

Digging further into the enumeration history shows two items were opened inside that archive: a folder named shard_references, and a file named shard_ref_011_ember_court_piece.txt, both timestamped 2026-06-18 13:18:23Z , six minutes before shardchain.tar was opened at 2026-06-18 13:24:55Z:

```
RecentDocs
**All values printed in MRUList\MRUListEx order.
Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs
LastWrite Time: 2026-06-18 13:42:24Z
  16 = KAPE
  15 = Capture
  14 = shardchain.tar
  13 = shard_references
  12 = shard_ref_011_ember_court_piece.txt
  11 = Personal
  10 = marr_working_notes_lowtide.txt
  9 = Threat/
  8 = ms-gamingoverlay:///
  7 = kglcheck/
  6 = Desktop
  5 = current_working
  0 = Registry
  4 = internal_reports
  3 = shard_ref
  2 = custody_chains
  1 = Documents

Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.tar
LastWrite Time 2026-06-18 13:24:55Z
MRUListEx = 0
  0 = shardchain.tar

Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\.txt
LastWrite Time 2026-06-18 13:18:23Z
MRUListEx = 1,0
  1 = shard_ref_011_ember_court_piece.txt
  0 = marr_working_notes_lowtide.txt

Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs\Folder
LastWrite Time 2026-06-18 13:18:23Z
MRUListEx = 1,0
  1 = shard_references
  0 = Personal
```

Since shard_references sits inside the same Documents\Registry\ tree as oath_records_cinderbound_vol2.zip, tracing the full path shows that the deepest folder enumerated within the archive is saltoaths_secretive.

> **Answer:** saltoaths_secretive

Flag 4. Staging Location for Stolen Records

> ***Question:*** *Before exfiltration comes collection, files pulled from their places and gathered where the operative controls. Where did CROWQUILL stage the stolen records?*

Methodology

The copy history registry artifact shows a set of files being moved to a common destination:

```
FM\CopyHistory:
  c:\users\public\music\saltwork\
  c:\users\public\music\saltwork
```

Folder-access history corroborates this, showing that the same path, C:\Users\Public\Music\saltwork\ , was opened in Windows Explorer:

```
FM\FolderHistory:
  c:\users\vmarr\desktop\working\
  c:\users\public\music\saltwork\
  c:\users\vmarr\appdata\Roaming\KeePass\
  c:\users\vmarr\appdata\Roaming\
  c:\users\vmarr\appdata\
  c:\users\vmarr\appdata\Local\
  C:\Users\vmarr\Documents\Registry\shard_references\
  C:\Users\vmarr\Documents\Registry\
  C:\Users\vmarr\Documents\Registry\shard_storage\
  C:\Users\vmarr\Documents\Registry\shard_storage\ShardKeepass_FirstMark\
  C:\Users\vmarr\Documents\Registry\shard_ref\
  C:\Users\vmarr\Documents\Registry\internal_reports\
  C:\Users\vmarr\Documents\Registry\custody_chains\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\oath_records_cinderbound_vol2\
  C:\Users\vmarr\Documents\Registry\oath_records_cinderbound_vol2.zip\oath_records_cinderbound_vol2\saltoaths_secret
ive\
  C:\Users\vmarr\Documents\
  C:\Users\vmarr\Documents\Personal\
  C:\Users\vmarr\
```

Finally, the TypedPaths registry key shows the same path was manually typed into Explorer's address bar, confirming it as the staging directory:

```
----------------------------------------
typedpaths v.20200526
(NTUSER.DAT) Gets contents of user's typedpaths key

Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths
LastWrite Time 2026-06-18 13:20:32Z

url1     C:\Users\Public\Music\saltwork
----------------------------------------
```

> **Answer: C:\Users\Public\Music\saltwork**

Flag 5. Compressed File Prior to Exfiltration

> ***Question:*** *Based on the compression history, what file was compressed prior to exfiltration?*

Methodology

The compression history recorded via the 7-Zip plugin shows exactly one file being compressed on the system:

```
Compression\ArcHistory:
  C:\Users\Public\Pictures\shardchain.tar

Extraction\PathHistory:
```

> **Answer: C:\Users\Public\Pictures\shardchain.tar**

Flag 6. Location of the Master Credential Store

> ***Question:*** *One file above all others, holding keys to every shard, every custodian, every oath. Whoever holds this holds the right to reconstruct authority older than any crown. Where was it stored?*

Methodology

Continuing the same folder-enumeration trail from Flag 3 and Flag 4, the KeePass database referenced throughout the earlier artifacts resolves to a single path under vmarr's Documents\Registry tree.

> **Answer: C:\Users\vmarr\Documents\Registry\shard_storage\ShardKeepass_FirstMark**

Flag 7. Final Folder of Operation

> ***Question:*** *When staging was complete and the archive sealed, the trail ends in one folder — where enumeration stopped. Where did CROWQUILL conclude their operations while using 7-Zip?*

Methodology

The last folder referenced in the 7-Zip enumeration history, where activity stops after the archive was sealed, is vmarr's desktop working directory.

> **Answer: C:\Users\vmarr\Desktop\working**

Summary

Working purely from Windows Registry artifacts via RegRipper, the investigation traced CROWQUILL's full attack path: deploying KeeFarce to dump credentials from a live KeePass process, enumerating an archive of sensitive "oath" records, staging the stolen files under Public\Music\saltwork, compressing them into shardchain.tar, and finishing operations from a working directory on the desktop. This demonstrates how registry-based artifacts alone, without any file-system or memory image, can reconstruct a complete timeline of attacker activity.